

(d) (i) The main program needs extending to:

- output the content of the unsorted array using `OutputArray()`
- sort the array using `InsertionSort()`
- output the content of the sorted array using `OutputArray()`

Write program code to extend the main program.

Save your program.

Copy and paste the program code into part **2(d)(i)** in the evidence document.

[2]

(ii) Test your program.

Take a screenshot of the output(s).

Save your program.

Copy and paste the screenshot into part **2(d)(ii)** in the evidence document.

[1]

(e) The function `Search()` performs a binary search to find `ItemToFind` in `DataArray`

The function takes **two** parameters:

- `DataArray`, an array of integers
- `ItemToFind`, an integer to find in `DataArray`

The function returns:

- the index of `ItemToFind` if it is in `DataArray`
- `-1` if `ItemToFind` is **not** in `DataArray`

Write program code for `Search()`

You must **not** use any inbuilt searching functions for your programming language.

Save your program.

Copy and paste the program code into part **2(e)** in the evidence document.

[6]